CHAITANYA MENON

Roll No.: 231DS005
BTech in Computational and Data Science
Minor in Management
National Institute of Technology, Karnataka

+91-8714727341
chaitanyamenon2005@gmail.com
chaitanya.231ds005@nitk.edu.in
chaitanyamenon0902
chaitanya-menon

Education

BTech in Computational and Data Science *2023-Present*
National Institute of Technology Karnataka CGPA: 7.71

Senior Secondary *2020-2022*
Narayana PU College Percentage: 94

Secondary *2012-2020*
Sharada Vidya Mandira Percentage: 94.6

Projects

AlgoArena: Collaborative Coding Platform *Nov. 2025-Mar. 2026*
Tools: React.js,Node.js,Express,MongoDB,Websocket

- Built live collaborative coding environment used by 50+ users simultaneously, cutting setup time by 40%
- Implemented JWT authentication system handling 100+ daily active users with 99.8% uptime
- Built room management system supporting 5 programming languages with 200ms synchronization latency

Spotify Genre Prediction: Predict Genre of Artist's next top hit *Sept. 2025-Dec. 2025*
Tools: Postman, Spotify API, TenserFlow,Scikit-Learn,Matplotlib

- Achieved 85% accuracy in genre classification using Random Forest on 10,000+ song samples
- Implemented SMOTE to address class imbalance and enhance model performance on minority genres
- Engineered a recommendation module that maps artist audio features to likely genres
- Performed data preprocessing and feature engineering on a structured dataset containing 15+ audio attributes

Potato Disease Detection using CNN *Dec. 2025- Jan. 2026*
Tools: TensorFlow, React, GCP, Python

- Trained a CNN using TensorFlow to classify potato plant diseases—with 97% test accuracy with 3000+ images
- Reduced false positives by 18% via architecture tuning and image augmentation techniques.
- Deployed TensorFlow model on Google Cloud Platform with React frontend for real-time potato disease detection

ISTE APP: An App exclusively for ISTE members *Nov. 2025-Mar. 2026*
Tools: Firebase, Flutter, Android Studio

- Developed a mobile application using Flutter, enabling event browsing, participation tracking and user profiles
- Integrated Firebase Authentication with role-based access and password-protected event invites
- Designed data models to manage events, calendar scheduling, and participation records with real-time updates

Technical Skills and Interests

Languages: C++,Python,HTML,SQL
Web Technologies: React, Node.js, Express, MongoDB, Prisma, Firebase
ML and Data Analysis: TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Tools: Git, Power BI, Tableau, GCP, AWS
Coursework: Database Management System, Operating System, Data Structures and Algorithms, Computational Linear Algebra, Object Oriented Programming, Machine Learning, Computer Architecture, Probability, Time Series
Areas of Expertise: Machine Learning, Web Development, Data Science and Analytics, Full Stack Development, Visualization and Dashboards

Positions of Responsibility

- **Website Committee Head (WebMaster)**, ISTE NITK
  Lead the development, deployment and maintenance of the website using React (Vite) and Prisma with PostgreSQL
- **Rotaract Member**, Pragati Initiative
  Volunteered weekly to mentor and teach underprivileged school students in mathematics and foundational subjects

Achievements

- **AWS Cloud Practitioner Certification**
  Completed foundational training on core AWS services and cloud architecture.
- **NPTEL: Introduction to Large Language Models (LLMs) — Top 5% Nationwide**
  Covered foundational concepts in LLM architecture, training, fine-tuning, deployment and GenAI applications.
- **Saras AI Institute: CLASH-OF-T-AI-TANS (Hackathon)**
  Placed among top 50 teams in an AI-focused hackathon to develop innovative machine learning solutions.